

Assignment-3: Quantitative Finance Lab

Optimization Algorithms & Results Documentation

Q1: Gradient-Based Optimization with Restarts

Description:

Implements gradient-based optimization (L-BFGS-B) with multiple random restarts for three benchmark functions:

- Rastrigin
- Rosenbrock
- Himmelblau

Results (Q1.csv):

Function	Best	Worst	Average	Std. Dev.
Rastrigin	7.11e-15	1.99	0.895	0.470
Rosenbrock	3.12e-14	6.13e-12	1.90e-12	1.66e-12
Himmelblau	1.06e-17	1.15e-15	4.90e-16	3.28e-16

Explanation:

The optimizer reliably finds the global minimum for all three functions (values near zero), with very low variance, indicating robust performance across random initializations.

Q2: Binary Genetic Algorithm (GA) for Optimization

Description:

Implements a binary-coded GA for the same benchmark functions. Results are compared to gradient-based optimization.

Results (BinaryGA.csv):

Function	Best	Worst	Average	Std. Dev.
Rastrigin	2.42e-06	2.24	0.647	0.691

Function	Best	Worst	Average	Std. Dev.
Rosenbrock	3.28e-04	5.06	0.535	0.987
Himmelblau	7.75e-08	0.989	0.0796	0.211

Explanation:

The binary GA finds near-optimal solutions but with higher variance and slightly worse average performance compared to gradient-based methods, especially for more complex landscapes.

Q3: Parameter Sensitivity Analysis for GA

Description:

Analyzes how mutation and crossover probabilities affect GA performance on Himmelblau's function.

Results (ParaSensitiveRes.csv):

muteProb	crossProb	AvgFit	StdDev	BestFitness	WorstFit
0.01	0.7	0.161	0.379	1.41e-06	1.45
0.01	0.8	0.0685	0.267	7.75e-08	1.48
0.01	0.9	0.0731	0.196	1.74e-07	1.01
0.02	0.7	0.0533	0.265	7.75e-08	1.48
0.02	0.8	0.0465	0.138	7.75e-08	0.666
0.02	0.9	0.0742	0.281	7.75e-08	1.45
0.05	0.7	0.0386	0.133	7.75e-08	0.666
0.05	0.8	0.000486	0.00194	7.75e-08	0.0107
0.05	0.9	0.000356	0.00191	7.75e-08	0.0107

Explanation:

Lower mutation and higher crossover probabilities generally yield better average fitness. The best results are achieved with moderate mutation (0.05) and high crossover (0.8–0.9).

Q4: Comparison of Binary GA and Real-Coded GA

Description:

Compares the performance of Binary GA and Real-Coded GA (using SBX crossover and polynomial mutation) on Himmelblau's function.

Results (GAComparision.csv):

	Best	Worst	Average	Std. Dev.	Median
Binary GA	7.75e-08	2.88	0.130	0.525	0.000602
Real GA	1.89e-08	0.000163	1.69e-05	3.06e-05	4.99e-06

Explanation:

The Real-Coded GA consistently outperforms the Binary GA, achieving lower best, average, and median fitness values, and much lower variance.

Q5: Constrained Optimization (Himmelblau with Constraints)

Description:

Solves Himmelblau's function with two circular constraints using both Binary GA and Real-Coded GA, employing a parameter-free penalty method.

Results (GAResults.csv):

	Best	Worst	Average	Std. Dev.	FeasibleRuns	FeasiblePercent
Binary GA	13.591	13.591	13.591	0.0	1.0	3.33
Real GA	13.595	13.595	13.595	0.0	1.0	3.33

Explanation:

Both algorithms found feasible solutions in only 1 out of 30 runs (3.33%), with similar best and average fitness. The constrained problem is significantly harder, and the feasible region is small and crescent-shaped.

Visualizations

- Each question folder contains PNG plots visualizing convergence, boxplots, parameter sensitivity, and feasible regions.
 - These plots help illustrate algorithm performance, solution quality, and the effect of parameter choices.
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Summary

- **Gradient-based optimization** is most reliable for unconstrained problems.
- **Binary GA** works but is less precise and more variable.
- **Real-Coded GA** (SBX + Polynomial Mutation) is superior to Binary GA for continuous problems.
- **Parameter sensitivity** analysis shows that moderate mutation and high crossover rates are optimal.
- **Constrained optimization** is challenging; both GAs struggle to find feasible solutions, highlighting the difficulty of the feasible region.